C73-27: exact finite-player selection of a nonminimizing potential equilibrium

Ji Ho Bae
LawAll AI

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Abstract

Fixed smooth three-state data give bounded symmetric exact Markov-perfect feedbacks for every population size. Their empirical paths converge to a constant equilibrium that does not minimize the potential action. We verify every continuation configuration and all bounded progressively measurable unilateral deviations, prove the full path limit, and exhibit a strictly better feasible path.

1 Statement and smooth data

Write $\Delta_3 = \{m \in [0,1]^3 : \sum_i m_i = 1\}$ and fix $T = 1$, $m_0 = (1/3, 1/3, 1/3)$. For the potential action in C73-27,

$$J(m,a) = \int_0^1 \left[\frac{1}{2} \sum_{i,j \neq i} m_i a_{ij}^2 + \mathcal{F}(m) \right] dt + \mathcal{G}(m(1)), \qquad \dot{m}_i = \sum_{j \neq i} (m_j a_{ji} - m_i a_{ij}), \tag{1}$$

the admissible paths are absolutely continuous simplex paths starting at m_0 with measurable nonnegative off-diagonal rates and finite weighted quadratic energy. Denote the paths in global minimizers by $\mathcal{M}_{1,m_0}$. In the finite game, player ℓ's terminal cost is $g_{X_1^\ell}(m_1^{N,-\ell})$, where $g_i = \partial_i \mathcal{G}$ and $m^{N,-\ell}$ is the empirical law of the other $N-1$ players. The running cost is $\frac{1}{2} \sum_{j \neq X_t^\ell} (a^\ell_{X_t^\ell j})^2 + f_{X_t^\ell}(m_t^{N,-\ell})$, with $f_i = \partial_i \mathcal{F}$.

Theorem 1. *There are C^∞ potentials on $\mathbb{R}^3$ and, for every $N \geq 2$, bounded symmetric Markov-perfect feedbacks for exactly these games whose empirical processes satisfy*

$$\sup_{0 \leq t \leq 1} \|\mu_t^N - m_0\|_2 \longrightarrow 0 \quad \textit{in probability}, \qquad \mathcal{L}(\mu^N) \Longrightarrow Q = \delta_{\bar{m}} \quad \textit{in } D([0,1], \Delta_3) \textit{ with } J_1,$$

where $\bar{m}(t) = m_0$. This continuous limit has $Q(\mathcal{M}_{1,m_0}) = 0$. In particular, the universal selection assertion in the stated C73-27 formulation is false.

Set $a = 2/5$, $b = 3/5$ and define

$$\rho(r) = \begin{cases} e^{-1/r}, & r > 0, \\ 0, & r \leq 0, \end{cases} \qquad \psi(z) = \frac{\rho(z-a)}{\rho(z-a) + \rho(b-z)}.$$

The denominator is positive everywhere. Every right derivative of $e^{-1/r}$ is a polynomial in $1/r$ times $e^{-1/r}$ and tends to zero at 0, so ρ and ψ are smooth. The quotient rule gives

$$\psi'(z) = \frac{\rho'(z-a)\rho(b-z) + \rho(z-a)\rho'(b-z)}{[\rho(z-a) + \rho(b-z)]^2} \geq 0.$$

Thus $\psi = 0$ on $(-\infty, a]$, $\psi = 1$ on $[b, \infty)$, and $L := \sup_{\mathbb{R}} \psi' < \infty$. With $S(m) = \sum_i m_i$, choose

$$\mathcal{F}(m) = 0, \qquad \mathcal{G}(m) = -\psi(m_1) + (S(m) - 1)\psi'(m_1). \tag{2}$$

These are fixed independently of N. Direct ambient differentiation yields $\partial_1\mathcal{G} = (S-1)\psi''(m_1)$ and $\partial_2\mathcal{G} = \partial_3\mathcal{G} = \psi'(m_1)$. Consequently on the simplex the actual data are

$$\mathcal{G}(m) = -\psi(m_1), \qquad (g_1, g_2, g_3) = (0, \psi'(m_1), \psi'(m_1)), \qquad f_i = 0. \tag{3}$$

All individual terminal costs lie in $[0, L]$.

2 An exact feedback for every finite game

For $x \in \{1,2,3\}^N$ let $k_\ell(x) = \#\{r \neq \ell : x_r = 1\}$ and write $x^{\ell\to j}$ for the vector with coordinate ℓ replaced by j. Construct scalar functions in reverse time $\tau \in [0,1]$ by

$$w'_{N-1} = -\tfrac{1}{2}w_{N-1}^2, \qquad w_{N-1}(0) = \psi'(1) = 0, \tag{4}$$

$$w'_k = (N-1-k)w_k w_{k+1} - (N-k-\tfrac{1}{2})w_k^2, \qquad w_k(0) = \psi'(k/(N-1)), \quad 0 \le k < N-1. \tag{5}$$

Lemma 1. *This system has a unique C^1 solution with $0 \le w_k \le L$. If $\psi'(k/(N-1)) = 0$, then $w_k \equiv 0$.*

Proof. The top function is zero. Proceed downwards, assuming $0 \le w_{k+1} \le L$. Put $A_k = (N-1-k)w_{k+1}$, $B_k = N-k-1/2$ and $c_k = w_k(0)$. The equation $w'_k = A_k w_k - B_k w_k^2$ is locally Lipschitz in w_k. For $c_k = 0$ its unique solution is zero. For $c_k > 0$ its global solution is

$$w_k(\tau) = \frac{c_k \exp(\int_0^\tau A_k(r)dr)}{1 + B_k c_k \int_0^\tau \exp(\int_0^s A_k(r)dr)ds}.$$

Differentiation verifies the equation and initial value. The denominator is positive. Whenever $w_k > L$, its derivative is bounded above by $(N-1-k)Lw_k - (N-k-1/2)w_k^2 < 0$. A connected interval above L would therefore be strictly decreasing from its entry value L, a contradiction. This proves the bound and the induction. □

Define values and outgoing feedbacks by

$$v_\ell(t,x) = \begin{cases} 0, & x_\ell = 1, \\ w_{k_\ell(x)}(1-t), & x_\ell \in \{2,3\}, \end{cases}$$
$$q_j^\ell(t,x) = \begin{cases} w_{k_\ell(x)}(1-t), & x_\ell \in \{2,3\},\ j = 1, \\ 0, & \text{otherwise}, \end{cases} \qquad j \neq x_\ell. \tag{6}$$

The diagonal is minus the outgoing sum. Rows for counterfactual own states are given by the same rule after replacing x_ℓ by that state, so these define admissible full rate matrices. They are continuous in time, bounded by L independently of N, and symmetric under player permutations. Independent Poisson transition noises implement the joint chain; its total intensity is at most NL, so it is nonexplosive and uses no common noise.

The exact Nash equations are

$$\partial_t v_\ell + \sum_{r\neq\ell}\sum_{j\neq x_r} q_j^r[v_\ell(t,x^{r\to j}) - v_\ell(t,x)] - \tfrac{1}{2}\sum_{j\neq x_\ell}[v_\ell(t,x) - v_\ell(t,x^{\ell\to j})]_+^2 = 0. \tag{7}$$

Their terminal condition is exactly $v_\ell(1,x) = g_{x_\ell}(m_x^{N,-\ell})$. To verify (7) at every configuration, first suppose $x_\ell = 1$. Its value is zero under all opponent changes, and own alternative values are nonnegative. Thus all terms vanish and its optimal rates are zero. If $x_\ell \neq 1$, set $k = k_\ell(x)$ and $\tau = 1-t$. Its own jump to 1 changes its value from w_k to 0; switching between 2 and 3 leaves its value unchanged. Its own Hamiltonian is therefore $w_k^2/2$. Of the opponents, the k in state 1 have zero rates. Each of the $N-1-k$ others has exactly k state-1 opponents and jumps to 1 at rate w_k, changing the focal value to w_{k+1}. Equation (7) becomes $-w'_k + (N-1-k)w_k(w_{k+1}-w_k) - w_k^2/2 = 0$, precisely (5); when $k = N-1$ it is (4). In all cases,

$$q_j^\ell(t,x) = [v_\ell(t,x) - v_\ell(t,x^{\ell\to j})]_+. \tag{8}$$

Proposition 1. *The feedbacks* (6) *are exact Markov-perfect Nash equilibria against every bounded progressively measurable unilateral rate deviation.*

Proof. Fix any initial continuation time t_0, state vector x and player ℓ, retaining the opponents' feedbacks. Let its outgoing rates $\alpha_j(t) \geq 0$ be any admissible bounded progressively measurable deviation, possibly depending on the entire history, and let X be the resulting state process. Put $\Delta_j = v_\ell(t, X_t^{\ell\to j}) - v_\ell(t, X_t)$. The compensated-jump formula has a mean-zero martingale because the values, their time derivatives for fixed N, and all intensities are bounded. Using (7) in that formula gives exactly

$$\begin{aligned}\mathbb{E}\left[\int_{t_0}^1 \tfrac{1}{2}\sum_{j\neq X_t^\ell}\alpha_j(t)^2 dt + g_{X_1^\ell}(m_1^{N,-\ell})\right] - v_\ell(t_0,x)\\ = \mathbb{E}\int_{t_0}^1 \sum_{j\neq X_t^\ell}\left[\tfrac{1}{2}(\alpha_j - q_j^\ell)^2 + \alpha_j(\Delta_j)_+\right] dt \geq 0. \end{aligned} \tag{9}$$

Indeed, for $\alpha \geq 0$ and $q = (-\Delta)_+$, $\alpha^2/2 + \alpha\Delta + q^2/2 = (\alpha - q)^2/2 + \alpha\Delta_+$. By (8) the prescribed feedback makes this remainder zero. It therefore attains the minimum against all allowed deviations. The choices of t_0, x, ℓ were arbitrary, proving the continuation condition at every time and configuration, including rare states. □

3 Convergence of the entire empirical path

Take the required independent initial states with law m_0, independent of the transition noises. Let K_N count those initially in state 1, and set $A_N = \{K_N \leq (2/5)(N-1)\}$. On A_N, every player outside state 1 has K_N state-1 opponents, so its initial scalar terminal datum is zero. Lemma 1 gives $w_{K_N}(1-t) = 0$ at every time. Players in state 1 also have zero rates. This configuration is absorbing under the exact feedback, and $\mu_t^N = \mu_0^N$ throughout $[0,1]$ on A_N.

Here K_N is binomial with mean $N/3$ and variance $2N/9$. For $N \geq 12$ the threshold exceeds its mean by at least $N/30$; Chebyshev's inequality gives $\mathbb{P}(A_N^c) \leq 200/N$. Also $\mathbb{E}\|\mu_0^N - m_0\|_2^2 = 2/(3N)$. Since the squared simplex diameter is at most 2,

$$\mathbb{E}\sup_{t\leq 1}\|\mu_t^N - m_0\|_2^2 \leq \frac{2}{3N} + 2\mathbb{P}(A_N^c) \leq \frac{2}{3N} + \frac{400}{N} \longrightarrow 0. \tag{10}$$

Markov's inequality proves uniform convergence in probability. Choosing the identity time change bounds the J_1 distance by the uniform distance, so the laws converge in the J_1 topology to $Q = \delta_{\bar{m}}$. The limiting path is continuous.

4 Existence of minimizers and strict nonminimality

For completeness, the minimizing-path set is nonempty and Borel. Introduce fluxes $z_{ij} = m_i a_{ij} \geq 0$. The constraints become linear in z, and the energy is $K(m,z) = \frac{1}{2}\int_0^1 \sum z_{ij}^2/m_i\, dt$, with $0/0 = 0$ and $z^2/0 = +\infty$ for $z > 0$. The constant zero-flux path is feasible. Since the terminal potential belongs to $[-1, 0]$, a minimizing sequence has bounded energy, hence bounded L^2 fluxes because $m_i \leq 1$. The forward equation gives bounded L^2 path derivatives and a common 1/2-Hölder modulus. Arzelà–Ascoli and weak L^2 compactness give uniform $m^n \to m$ and weak $z^n \rightharpoonup z$ along a subsequence. Nonnegative test functions preserve $z \geq 0$, and time interval indicators pass the integrated constraints to the limit.

The exact dual representation

$$K(m,z) = \sup_{q\in L^\infty}\int_0^1 \sum_{i,j\neq i}[q_{ij}z_{ij} - \tfrac{1}{2}m_i q_{ij}^2]dt$$

proves lower semicontinuity: each bounded q gives a continuous expression under these convergences. Equality in the representation follows by the bounded choices $q_{ij} = \min\{A, z_{ij}/m_i\}$, assigning A where $m_i = 0 < z_{ij}$ and zero where both vanish, and monotone convergence as $A \uparrow \infty$. Continuity passes the terminal potential to the limit. Thus the limit minimizes. Finite energy forces $z_{ij} = 0$ almost everywhere on $\{m_i = 0\}$; defining $a_{ij} = z_{ij}/m_i$ on $\{m_i > 0\}$ and zero elsewhere recovers admissible measurable rates. The same argument for any sequence of minimizers shows that their law paths form a uniformly compact set. Its inclusion in D is continuous for J_1, so it is compact and Borel there as well.

Every rate representation of $\bar{m}$ has nonnegative action, since $\mathcal{F} = 0$, $\mathcal{G}(m_0) = 0$, and kinetic energy is nonnegative. For a competitor choose $c = \log 2$, $a_{21} = a_{31} = c$, and all other off-diagonal rates zero. The forward law is

$$m_1(t) = 1 - \tfrac{2}{3}e^{-ct}, \qquad m_2(t) = m_3(t) = \tfrac{1}{3}e^{-ct}.$$

It is admissible and has terminal law $(2/3, 1/6, 1/6)$, where $\mathcal{G} = -1$. Its energy is $\frac{1}{2}\int_0^1 (m_2 + m_3)c^2 dt = c(1 - e^{-c})/3 = (\log 2)/6$. Therefore

$$J(m, a) = \frac{\log 2}{6} - 1 < 0. \tag{11}$$

For example $\log 2 = \int_1^2 x^{-1} dx < 1$ proves this strict inequality. The global minimum is negative, so $\bar{m}$ cannot occur in any minimizing pair. This gives $Q(\mathcal{M}_{1,m_0}) = 0$ and completes Theorem 1.

5 The prescribed representatives and source scope

The normal term in (2) vanishes on the simplex but adds the common component $\psi'(m_1)$ to the ambient gradient of $-\psi(m_1)$. Its tangent projection is $(-2\psi'/3, \psi'/3, \psi'/3)$, exactly the intrinsic gradient of the restricted potential. This is allowed by the ambient derivative prescription in C73-27 and by the projected potential convention in [2, Eq. (2.9)]; no hypothesis requires $\sum_i g_i = 0$. We retain the actual costs (3) in every finite continuation game. State-dependent common additions are not asserted to preserve finite closed-loop feedbacks. Centering those costs would change the finite game and would require another analysis.

The published two-state selection result analyzes a specified anti-monotone model, rather than all smooth terminal costs: [3, Section 3] uses $G(x, m) = -mx$ on $\{-1, 1\}$ with zero interaction running cost. The potential selection discussion in [1, Section 4.3] supplies the motivation; the theorem here addresses the catalogue formulation.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

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